Lab 1: Identify Entities and Attributes from a Flat File

1. Order
    a. order_id: 1001
    b. orded_date: 2026-06-01
2. Customer
    a. customer_name: Jane Doe
    b. customer_email: jane@example.com
    c. shipping_address: 123 Elm St
3. Product
    a. product_id: 200
    b. product_name: Widget A
    c. price: 19.99
4. OrderItem
    a. quantity: 2

Lab 2: Classify Relationships and Keys

Entity 1	Entity 2	Relationship
Customer	Order	1:N
Order	Product	M:N
Order	OrderItem	1:N
Product	OrderItem	1:N

Lab 3: Apply First Normal Form (1NF)

order id	customer	product_id	quantity
1002	Bob Smith	10	2
1002	Bob Smith	20	1

Lab 4: Convert Repeating Groups into Rows

```
[
    {'order_id': 1003, 'customer': 'Ana', 'product_id': 301, 'qty': 2},
    {'order_id': 1004, 'customer': 'Ben', 'product_id': 302, 'qty': 1},
    {'order_id': 1004, 'customer': 'Ben', 'product_id': 302, 'qty': 1}
]
```

Lab 5: Apply Second Normal Form (2NF)

Table 1 (orders):
Columns: order_id, qty, product_id, price

Table 2 (products):
Columns: product_id, product_name, product_price

Lab 6: Apply Third Normal Form (3NF)

Table 1 (customers):
Columns: customer_id (pk), customer_name, city_id(fk)

Table 2 (cities)
Columns: city_id(pk), city_name, zip, state_name

Lab 7: Design an OLTP Normalized Schema (Orders)

```
CREATE TABLE customers (
    customer_id INTEGER PRIMARY KEY,
    name TEXT NOT NULL,
    email TEXT NOT NULL UNIQUE,
    city TEXT,
    country TEXT
);

CREATE TABLE products (
    product_id INTEGER PRIMARY KEY,
    product_name TEXT NOT NULL,
    unit_price DECIMAL(10, 2) NOT NULL
);

CREATE TABLE orders (
    order_id INTEGER PRIMARY KEY,
    customer_id INTEGER NOT NULL,
    order_date DATE NOT NULL,
    FOREIGN KEY (customer_id) REFERENCES customers(customer_id)
);

CREATE TABLE order_items (
    order_item_id INTEGER PRIMARY KEY,
    order_id INTEGER NOT NULL,
    product_id INTEGER NOT NULL,
    quantity INTEGER NOT NULL,
    unit_price DECIMAL(10, 2) NOT NULL,
    FOREIGN KEY (order_id) REFERENCES orders(order_id),
```

```
    FOREIGN KEY (product_id) REFERENCES products(product_id)
);
```

Lab 8: Create a Denormalized Reporting Table

```
CREATE TABLE daily_orders_report (
    order_id INTEGER PRIMARY KEY,
    customer_name TEXT NOT NULL,
    order_date DATE NOT NULL,
    total_order_value DECIMAL(10, 2) NOT NULL,
    product_list TEXT,
    shipping_city TEXT
);

INSERT INTO daily_orders_report
     (order_id, customer_name, order_date, total_order_value,
product_list, shipping_city)
VALUES(1001, 'Tony', '2026-07-15', 87.50, 'Wireless Mouse, Clean
Code, Notebook', 'Berlin');
```

Lab 9: Design a Star Schema: Fact and Dimension Tables

FactOrders:
- order_id
- customer_id
- product_id
- qty
- price
- order_date

CustomerDim
- customer_id
- customer_name
- customer_email
- shipping_adresses

ProductDim
- product_id
- product_name
- unit_price
- category

DateDim
- day
- month
- year

- quarter